



AEGIS

Privacy-Preserving On-Device Combat Motion Intelligence

Explainable real-time motion awareness designed for Snapdragon-powered HP PCs

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The Problem: Latency, Fragility, and Surveillance Exposure

01 / REACTIVE REVIEW

Incident Review Happens After the Fact

In athletic coaching and facility safety, video is examined hours or days later. Supervisors lack real-time warning to intervene before injuries occur.

02 / CLOUD DEPENDENCE

High Latency & Single Point of Failure

Cloud video analytics streams video over public internet, introducing 200-800ms delays and failing completely during network interruptions.

03 / TEMPORAL BLINDNESS

Single-Frame Models Miss Motion Context

Static single-frame object detectors cannot distinguish a high-velocity punch from an open high-five or a stumble from a dynamic athletic slide.

04 / PRIVACY VULNERABILITY

Biometric Harvesting & Facial Exposure

Streaming surveillance footage to external cloud servers creates catastrophic risks of non-consensual biometric data scraping and surveillance abuses.

The Solution: Edge-Native Biomechanical Intelligence

LOCAL PROCESSING

100% On-Device Execution

Processes video frames entirely in local RAM on Snapdragon PCs. Video buffers are overwritten immediately.

ANONYMIZED SKELETONS

Human Pose Landmarks

Extracts 9 normalized 2D skeletal landmarks (wrists, shoulders, hips, ankles). Completely strips facial identity.

TEMPORAL WINDOWS

Multi-Frame Motion Context

Maintains a 12-frame sliding window to calculate true physical velocities, body scale growth rates, and torso ratios.

ACTION VOCABULARY

6 Biomechanical Patterns

Deterministically detects Neutral, Guard, Punch-like extension, Kick-like extension, Rapid Approach, and Fall postures.

TRANSPARENT METRICS

Explainable Telemetry

Outputs human-verifiable evidence strings (e.g., 'right arm reach 1.65x, speed 1.15/s') alongside bounded threat scores.

HUMAN-IN-THE-LOOP

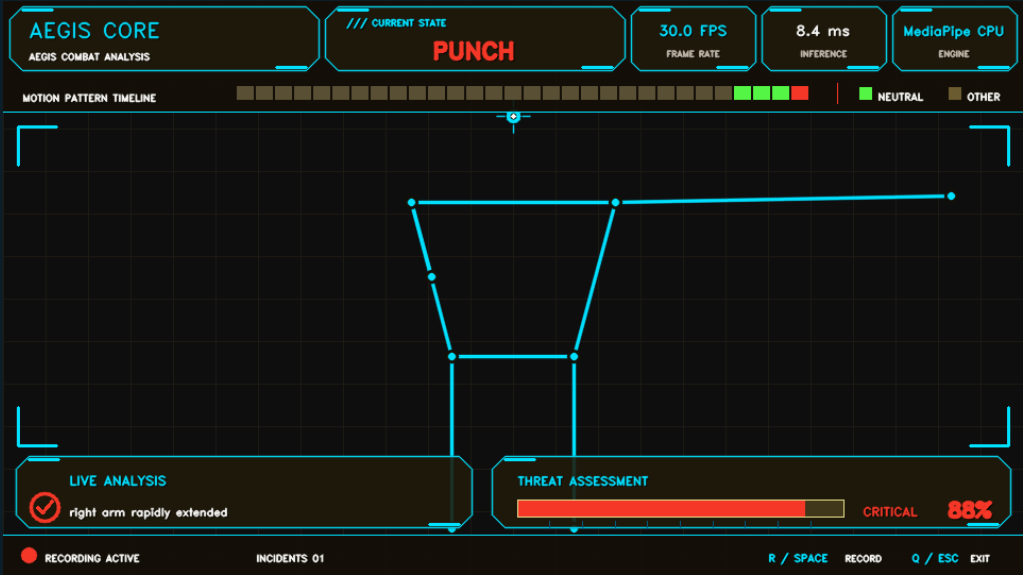
Decision-Support Telemetry

Empowers coaches and safety supervisors with visual gauges and opt-in local logging. Autonomous action is prohibited.

Technical Architecture: From Raw Pixels to Explainable Alerts

1. Optical Input:	Webcam / Prerecorded Video Clip
2. Normalization:	OpenCV Ingestion (960x540 Buffer)
3. Pose Estimator:	MediaPipe Pose (Active: MediaPipe CPU)
4. Landmark Array:	9 Normalized Keypoints (Visibility > 0.40)
5. Temporal Window:	12-Frame Deque Tracking Velocity
6. Pattern Engine:	Kinematic Rule Evaluator (Priority Order)
7. Threat Fusion:	Graduated Threat Index (0.0 - 1.0)
8. Output & Logs:	AEGIS CORE HUD & Local JSONL Log

AEGIS CORE TELEMETRY INTERFACE



Current Verified Engine: MediaPipe CPU (Active)

Live HUD running at 960x540 with FPS counter, inference latency, threat meter, and timeline.

Innovation, Privacy, and Responsible Safety Safeguards

ZERO FACIAL RECOGNITION

Biometric Privacy by Design

Aegis does not analyze facial landmarks or biometric templates. It is architecturally blind to human identity.

VOLATILE FRAME MEMORY

No Cloud Video Uploads

All computations execute in volatile RAM on the local workstation. Video streams are never sent to external servers.

OPT-IN LOCAL LOGGING

Operator-Controlled Storage

Incident logging is strictly opt-in via manual toggle (R or Space). Log records reside exclusively on the host PC in artifacts/.

INTENT VS. MOTION

Kinematic Signals, Not Guilt

Aegis measures biomechanical movement patterns; it does not and cannot infer criminal intent or moral culpability.

DECISION-SUPPORT ONLY

Prohibition of Autonomous Action

Outputs are engineered strictly for human review. Under no circumstances should Aegis trigger automated punitive actions.

SYNTHETIC EVALUATION

Camera-Free Demonstration

A deterministic synthetic demonstration mode (--demo and --export-preview) enables comprehensive validation without real users.

Qualcomm Snapdragon Optimization & NPU Deployment Path

[STAGE 1: FUNCTIONAL BASELINE]

Current MediaPipe CPU Pipeline

The validated MVP runs MediaPipe CPU to establish ground-truth algorithmic correctness, temporal stability, and deterministic test suites.

[STAGE 2: MODEL SELECTION]

Qualcomm AI Hub Model Catalog

Select an edge-optimized pose model (e.g., YOLOv8n-pose or RTMPose) from Qualcomm AI Hub with proven low-latency edge topology.

[STAGE 3: COMPILATION & QUANTIZATION] INT8 / FP16 Hexagon Compilation

Utilize Qualcomm AI Hub cloud toolchains to compile and quantize the model for the Hexagon NPU on Snapdragon X Elite/Plus platforms.

[STAGE 4: RUNTIME ADAPTATION]

ONNX Runtime QNN Provider

Integrate compiled ONNX assets via QNNExecutionProvider with Hexagon Tensor Processor delegates (QnnHtp.dll) behind SnapdragonSession.

[STAGE 5: RIGOROUS BENCHMARKING]

Identical-Video CPU vs QNN Benchmark

Execute scripts/benchmark.py on an identical 1080p clip to measure real throughput (FPS), p95 latency reduction, and CPU offload.

** Architectural Transparency Note: All QNN/NPU execution steps are documented as the planned optimization roadmap. Aegis currently executes and reports MediaPipe CPU.*

Evidence, Real-World Impact, and Project Roadmap

VERIFIED CURRENT EVIDENCE

- Functional Live Pipeline: Real-time webcam ingestion, tracking, threat scoring.
- 17 / 17 Tests Passing: Deterministic pytest suite verifying heuristics and safety bounds.
- Synthetic Demo Suite: Complete camera-free demo & preview image exporter.
- Truthful Telemetry: Truthfully reports MediaPipe CPU without fabricating NPU metrics.
- Public GitHub Repo: github.com/aniketkapgate7-crypto/Aegis-Fight-Pattern-Analyzer

DEVELOPMENT ROADMAP

- QNN Pose Model: Deploy Qualcomm AI Hub compiled YOLOv8-pose INT8 model to Hexagon NPU.
- Snapdragon PC Benchmarks: Evaluate throughput, latency, and power on Snapdragon X Elite laptops.
- Multi-Person Tracking: Extend kinematic engine to multi-subject spatial distance tracking.
- Curated Validation Set: Benchmark against controlled athletic sparring clips for recall/precision.

CLOSING VALUE STATEMENT

Edge-Native Motion Intelligence

Aegis proves that advanced physical motion intelligence can run fully on-device without compromising personal privacy, streaming raw video to cloud servers, or harvesting biometrics.

Designed for Snapdragon-powered HP PCs, Aegis showcases the transformative potential of edge NPU acceleration for human-in-the-loop safety and athletic intelligence.